

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I V. Sheersika (192472056) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Sheersika

19/20
25/11

1. Subnetting $192.168.20.0/24$.

Given, original network = $192.168.20.0/24$.

Lab A - /26

Subnet mask:

$$/26 = 255.255.255.192$$

$$\text{Block size} = 256 - 192 = 64$$

So each /26 Subnet contains 64 addresses.

Starting from $192.168.20.0$

Network address = $192.168.20.0$

First Host = $192.168.20.1$

Last Host = $192.168.20.62$.

Broadcast Address = $192.168.20.63$.

Usable hosts:

$$2^{32-26} - 2 = 2^6 - 2 = 64 - 2 = 62$$

Lab B - /27.

The next unused address after Lab A is

$192.168.20.64$.

Subnet mask: $/27 = 255.255.255.224$.

Block size : $256 - 224 = 32$.

Therefore,

Network address = $192.168.20.64$.

First Host = $192.168.20.65$

Last Host = $192.168.20.94$.

Broadcast Address : $192.168.20.95$

Usable hosts:

$$2^{32-27} - 2 = 2^5 - 2 = 32 - 2 = 30$$

Lab C - /28

The next unused address is 192.168.20.96

Subnet mask:

$$/28 = 255.255.255.240$$

Block size:

$$256 - 240 = 16$$

Therefore:

Network Address = 192.168.20.96

First Host = 192.168.20.97

Last Host = 192.168.20.110

Broadcast Address = 192.168.20.111

Usable hosts:

$$2^{32-28} - 2 = 2^4 - 2 = 16 - 2 = 14$$

2. Subnetting of Two Systems

Given:

System 1 = 192.168.1.130

System 2 = 192.168.1.190

Subnet mask = 255.255.255.192

This is a /26 mask.

Block size: $256 - 192 = 64$

Therefore, subnet ranges are:

0 - 63, 64 - 127, 128 - 191, 192 - 255

Both 130 & 190 fall in the 128-191 subnet

System 1 : 192.168.1.130

Subnet range:

192.168.1.128 - 192.168.1.191

Therefore:

Network Address = 192.168.1.128.

First Host = 192.168.1.129

Last Host = 192.168.1.190.

Broadcast Address = 192.168.1.191

Since:

$$129 \leq 130 \leq 190$$

192.168.1.130 is a valid host IP.

System 2: 192.168.1.190.

190 also belongs to the same 128-191 subnet.

Therefore:

Network Address = 192.168.1.128.

First Host = 192.168.1.129

Last Host = 192.168.1.190

Broadcast Address: 192.168.1.191

Since 190 is the last valid host address,

192.168.1.190 is a valid host IP

Usable hosts: $2^{32-26} - 2 = 2^6 - 2 = 62$

$\therefore$ Usable hosts per subnet = 62.

$\therefore$ Both systems belong to the same subnet

3. Subnetting 172.16.0.0/16

Given :

HR $\rightarrow$ 125

Sales $\rightarrow$ 126

IT $\rightarrow$ 127

Admin $\rightarrow$ 128

HR - 125

Subnet mask = 255.255.255.128

Total addresses:

$$2^{32-25} = 2^7 = 128$$

Usable :

$$128 - 2 = 126$$

Therefore:

Network = 172.16.0.0

Host range = 172.16.0.1 - 172.16.0.126

Broadcast = 172.16.0.127

Sales - 126

Next available address = 172.16.0.128

Total addresses: $2^{32-26} = 64$

Usable : $64 - 2 = 62$

Therefore:

Network = 172.16.0.128

Host range = 172.16.0.129 - 172.16.0.190

Broadcast = 172.16.0.191

17- 127

Next available address = 172.16.0.192

Total addresses: $2^{32-27} = 32$

Usable: $32 - 2 = 30$

Therefore:

Network = 172.16.0.192

Host range = 172.16.0.193 - 172.16.0.222

Broadcast = 172.16.0.223.

Admin - 128.

Next available address = 172.16.0.224

Total addresses: $2^{32-28} = 16$

Usable: $16 - 2 = 14$

Therefore:

~~Network = 172.16.0.224.~~

Host range = 172.16.0.225 - 172.16.0.238.

Broadcast = 172.16.0.239.

The last allocated address is 172.16.0.239

Therefore, the unused addresses begin from:
172.16.0.240.

The complete /16 block ends at: 172.16.255.255

4. Subnetting $172.16.10.0/24$ using 127

Original network = $172.16.10.0/24$.

New subnet = 127

No. of Subnets

Original bits = 24

New bits = 27

Borrowed bits : $27 - 24 = 3$.

No. of subnets : $2^3 = 8$ subnets.

No. of hosts

Host bits : $32 - 27 = 5$

Usable hosts : $2^5 - 2 = 32 - 2 = 30$

Find Subnet containing $172.16.10.78$

$/27$ subnet mask : $255.255.255.224$.

Block size : $256 - 224 = 32$

Subnet ranges are :

$0 - 31$, $32 - 63$, $64 - 95$, $96 - 127$, $128 - 159$, $160 - 191$, $192 - 223$,
 $224 - 255$.

$\Rightarrow 78$ falls b/w 64 & 95

Therefore,

Network address = $172.16.10.64$

First Host = $172.16.10.65$

Last Host = $172.16.10.94$

Broadcast Address = $172.16.10.95$

$\Rightarrow 172.16.10.78$ belongs to subnet $172.16.10.64/27$.

IPv6 Address Compression.

Given IPv6 address,

2001:0db8:0000:0000:0000:ff00:0042:8329

Remove leading zeros

IPv6 allows leading zeros in each group to be removed.

So

2001:db8:0:0:0:ff:00:42:8329

Replace consecutive zero groups

The three consecutive 0 groups can be replaced by ::.

Therefore the shortest form is

2001:db8::ff00:42:8329

Expand it back

Compressed:

2001:db8::ff00:42:8329.

Each group must contain four hexadecimal digits.

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

Find network prefix for 164.

A 164 means the first 64 bits represent the network.

Each IPv6 group contains 16 bits.

Therefore, $64/16 = 4$ groups.

first four groups:

2001:0db8:0000:0000

So the network prefix can be written as:

2001:db8:0:0::/64.

(or)

2001:db8::/64

No. of host addresses.

IPv6 contains 128 bits.

for /64: $128 - 64 = 64$ host bits.

No. of possible addresses: 64.

Unlike traditional IPv4 subnet calculations, we normally don't subtract 2 for network & broadcast addresses because IPv6 does not use a broadcast address.

②